Repeat Purchase Prediction

Pipeline Structure and Rationale

Group 22 • Machine Learning, NOVA IMS • dataset “Everything & Then Some” • Handout PDF

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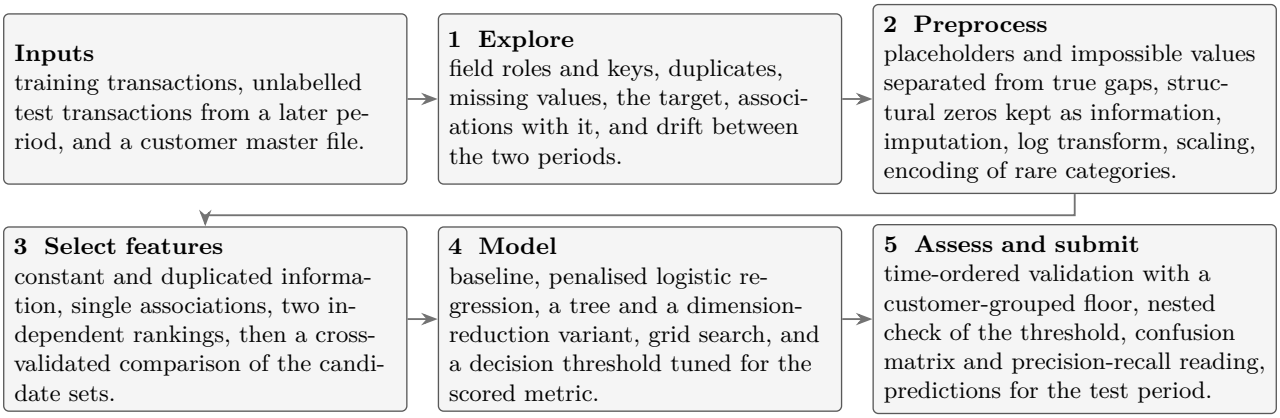
1. Objective and data

The task is to predict whether the customer behind a purchase will buy again within ninety days. Each row is one transaction, the answer is yes or no, and the competition scores the positive class with F1, so the work is judged on how reliably repeaters are found, not on how often the model is right overall.

Training covers 6,534 transactions from 610 customers between July 2015 and February 2025; the test file covers 986 transactions from 206 customers between May 2025 and October 2026 and carries no label. A customer file describes 1,143 accounts and joins on the customer key. The two transaction files share customers but not rows, and 94.5% of test rows belong to a customer the company has already seen, which is the most important fact about the evaluation. After cleaning, 56% of transactions repeat, so the classes are close in size and always answering yes already scores 0.718 on those rows.

2. Structure of the pipeline

Four notebooks, each handing its output to the next as a plain table.



3. The stages and why each one suits this data

Preprocessing. A placeholder for an unknown amount is not a large purchase, so those 35 cells and the 71 impossible negatives became missing values, flagged rather than clipped. An empty purchase history is not missing at all: 1,048 rows belong to customers with no earlier order, so those fields were set to zero and marked. Amounts are skewed, so a log transform and scaling follow, and rare categories were grouped before encoding.

Integrating the customer data. The customer file is optional, so it was joined only where it earns its place, and the key was checked first so no transaction is duplicated. Region and cohort carry real signal: one region repeats on 94.5% of its rows, almost 39 points above the average. Street, company name and postal code were left out: identifiers dressed as features, which would let the model memorise addresses instead of learning behaviour.

Feature selection. Selection was made by concept rather than by column, because encoding one category as several indicators makes those columns dependent. Constant and near-duplicate fields go first, then weak single associations, then two rankings that disagree in useful ways, and finally a cross-validated comparison on the time-ordered folds the model is scored on, preferring the smaller set. The funnel runs from 54 concepts and 125 columns to 18 and 63.

Modelling. With a few thousand rows of tabular, business-readable fields, a penalised logistic regression is the honest first choice: stable at this size, lightly tuned, and one interpretable coefficient per feature. A decision tree and a dimension-reduction variant were fitted as checks beside an always-yes baseline; the linear model won, 0.761 against 0.732 and 0.749.

Assessment. Every scored row is later than every row we can fit on, so the folds are cut in time order and none is fitted on the future; grouping them on the customer stays as the floor for the 5.5% of test

rows we have never met. The threshold was tuned on out-of-fold probabilities and re-verified inside a nested loop. A final check fits the earliest four fifths and scores the latest fifth.

4. Results

The time-ordered 0.761 is the headline, since no fold is ever fitted on the future, and the nested 0.759 says the tuned threshold adds nothing to it. The customer-grouped 0.740 is the floor for the 5.5% of test rows whose customer we have never met. The holdout on the latest fifth of the rows reaches 0.785, which is the realistic expectation because 94.5% of the test rows belong to a customer already seen, and the public part of the test set scored 0.795.

Measure	F1
Always-yes baseline, time-ordered folds	0.721
Time-ordered cross-validation	0.761
Customer-grouped cross-validation, reference model	0.740
Time-ordered holdout, latest fifth	0.785
returning customers / first appearance	0.793 / 0.710
Kaggle, public part of the test set	0.795

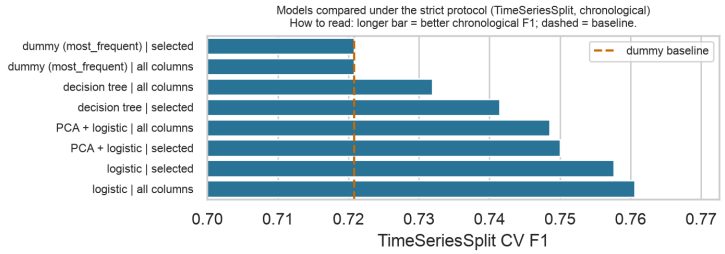


Figure 1. Every model family under the chronological protocol, on both feature sets, against the baseline. The bars sit close together: the honesty of the split matters more than the algorithm.

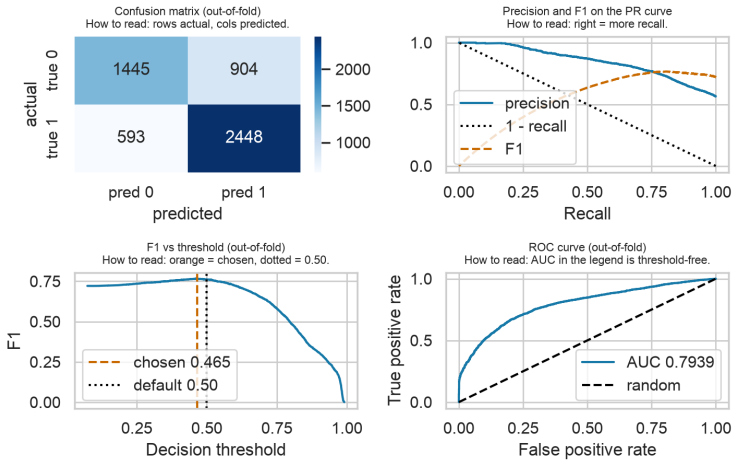


Figure 2. The assessment on the 5,390 rows a chronological split can score: confusion matrix at the chosen threshold, precision-recall trade, score against threshold, probability ranking alone.

Annexes

A. Reproducing the work. The four notebooks run in sequence in about a minute with a fixed random seed, and every stage writes its result as a readable table, so any number can be checked afterwards. Only tools covered in the course were used: no resampling, no bagging, no boosting.

B. Choices that were tested and then dropped. Customer-level history features, ratios, winsorising, quartile binning and interactions were all tried; none beat the noise between folds, and the customer aggregates made things worse. Scaling the yes-no flags as well as the amounts was also worse.

C. Limitations. The threshold was chosen on the training period and predicts more repeaters than that period holds, so it is the first value to revisit if the private leaderboard disagrees.